Confidence-Bounded Physical Feasibility Ranking for Neutronic Screening of QUBO-Feasible Fuel Layouts Using OpenMC

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***Abstract*—Constraint-solving and physical simulation expose different notions of feasibility, while reviewable decision-support workflows must preserve uncertainty and unresolved ties rather than force a unique winner. This paper presents Confidence-Bounded Physical Feasibility Ranking (CB-PFR), an auditable post-simulation decision rule for screening QUBO-feasible candidates against declared constraints using operational uncertainty bounds. The rule is specified through four testable properties: violations are non-decreasing in the bound multiplier; ranking and tie groups are invariant to input permutation; missing or non-finite required metrics fail closed; and at $z = 0$ the physical layer reduces to point-estimate feasibility-first ranking. We demonstrate the software workflow on a two-dimensional multigroup C5G7-derived fuel-placement case with four variable fuel positions, five fixed reflector positions, six feasible labels, and four diagonal-reflection symmetry classes. An 18-run exploratory campaign yields two passing and two failing classes under retrospective illustrative limits; a separate three-seed UMMU audit exposes unresolved dispersion and source-entropy-mesh concerns. The contribution is an evidence-auditable decision-support workflow, not a validated reactor-design method: thresholds are not engineering limits, equivalence is not established, and no physical optimum, calibrated confidence coverage, optimizer superiority, or quantum advantage is claimed.**

***Index Terms*—CB-PFR, C5G7 benchmark, OpenMC, QUBO, reproducible research software**

I. Introduction

A candidate may satisfy an encoded combinatorial constraint without satisfying a downstream physical criterion. Binary-quadratic objectives distinguish candidates only through represented terms [1], [2]. Reactor loading-pattern studies have explored genetic algorithms, simulated annealing, hybrid strategies, and problem-specific encodings [3]–[7]; more recent work combines adjoint-based physical information with evolutionary search [8]. These studies motivate separating the search representation from subsequent physical assessment, but their models and objectives differ and they are not direct performance baselines for this small enumerated case.

From an information-systems perspective, CB-PFR is framed here as a decision-support component: it exposes acceptance limits, metric uncertainty semantics, per-metric violations, missing-data states, and unresolved ties so that a user can audit why a candidate was screened in or out. This makes the decision pathway inspectable rather than treating a scalar score as a self-explanatory answer. At $z = 0$, the physical layer follows a point-estimate feasibility-first ordering related to Deb's constraint-handling rule [9]. For $z > 0$, the operational bounds can change decisions near a limit. This differs in purpose from stochastic ranking, which probabilistically balances objective value and constraint violation during evolutionary search [10]; CB-PFR is a post-simulation rule and this paper does not claim superiority over either method.

The paper makes three bounded contributions. First, it specifies four formal properties for an uncertainty-bounded feasibility and ranking layer and adds regression tests for those properties. Second, it separates substantive ranking scores from deterministic display order, allowing equivalent candidates to remain in the same tie group. Third, it audits a small C5G7-derived OpenMC dataset and documents how retrospective thresholds, inactive constraints, and unresolved source diagnostics affect interpretation. All six feasible layouts were enumerated; no pre-physics optimizer or annealer was run. Accordingly, the study evaluates the transparency and software behavior of the decision pathway, not optimization performance or physical validity.

II. Case Study and Methods

A. Candidate Space and Composition QUBO

The two-dimensional model fixes five reflector positions and varies four fuel positions $p_{00}, p_{01}, p_{10}, p_{11}$. Each fuel position receives either MOX or UO_2, with exactly two of each. The six label strings are MMUU, MUMU, MUUM, UMMU, UMUM, and UUMM. Let $y_i \in \{0, 1\}$ indicate MOX at position i. The composition-only penalty is

$$E(\mathbf{y}) = A\left(\sum_{i=1}^{4} y_i - 2\right)^2, \qquad A > 0. \tag{1}$$

Using $y_i^2 = y_i$, its expansion is

$$E(\mathbf{y}) = A\left(4 - 3\sum_i y_i + 2\sum_{i<j} y_i y_j\right). \tag{2}$$

Every feasible assignment has $\sum_i y_i = 2$, hence $E = 0$. This algebraic degeneracy is a limitation of the stated composition-only objective, not of QUBO in general. The study is a complete enumeration of a small design space and does not test whether a pre-physics optimizer is useful at larger scales.

B. Symmetry and Neutronic Metrics

The declared 3×3 position map places the four variable fuel locations at the upper-left 2×2 positions and the five reflectors elsewhere. Reflection across the main diagonal swaps p_{01} and p_{10}. Thus MMUU/MUMU form class S1, MUUM is S2, UMMU is S3, and UMUM/UUMM form S4. This class reduction depends on the documented mapping and symmetric model assumptions.

OpenMC is used for multigroup neutron-transport and tally calculations [11]; the C5G7 MOX specification provides the benchmark context [12]. For the four variable fuel regions, let R_i be the assembly-integrated fission tally. The relative peaking proxy is

$$F = \frac{\max(R_1, R_2, R_3, R_4)}{\frac{1}{4}\sum_{i=1}^{4} R_i}. \tag{3}$$

F is dimensionless and relative. It is not absolute power, pin power, a thermal-hydraulic quantity, or an approved safety criterion. Source-distribution entropy and k-generation traces are used as descriptive diagnostics; monitoring source convergence is important in eigenvalue calculations, but no single scalar diagnostic alone certifies convergence [13].

C. CB-PFR Rule

For candidate or symmetry class i and metric j, let $\bar{x}_{ij}$ be the observed estimate and s_{ij} the supplied uncertainty summary. CB-PFR forms operational bounds

$$B_{ij}^{-} = \bar{x}_{ij} - z_j s_{ij}, \quad B_{ij}^{+} = \bar{x}_{ij} + z_j s_{ij}. \tag{4}$$

For declared lower and upper limits L_j, U_j, violation is

$$v_{ij} = \max(0, L_j - B_{ij}^{-}) + \max(0, B_{ij}^{+} - U_j), \tag{5}$$

including only terms for limits that are declared. The normalized violation is $V_i = \sum_j w_j v_{ij}/d_j$, where $w_j \geq 0$ and $d_j > 0$ are the metric weight and scale. These operational bounds are not calibrated confidence intervals. Missing or invalid required values fail closed.

The substantive ranking key is lexicographic: QUBO-feasibility indicator, physical-feasibility indicator, weighted normalized physical violation, then QUBO energy. Candidate identifier may be used solely to make display order deterministic; it is excluded from the substantive key and cannot split a tie. A separate downstream selector may apply an explicitly declared secondary score, but it does not make the composition-only pre-physics QUBO informative and is not an optimizer-advantage result.

D. Formal Properties and Regression Tests

Let $s_{ij} \geq 0$ be the supplied uncertainty summary for candidate i and metric j, with fixed declared limits. The implementation and unit tests target the following properties.

P1 (monotone violation). For $z \geq 0$, each metric violation $v_{ij}(z)$ is non-decreasing in z. Indeed, $B_{ij}^{-} = \bar{x}_{ij} - z s_{ij}$ is non-increasing and $B_{ij}^{+} = \bar{x}_{ij} + z s_{ij}$ is non-decreasing; each positive-part violation term therefore cannot decrease as z grows. Thus the pass set can only stay unchanged or shrink.

P2 (permutation-invariant ranks and ties). For fixed candidate identifiers, estimates, and model, permuting the input sequence does not change candidate order after canonical display sorting or the membership of tie groups. The implementation excludes input index from the substantive ranking key; the stable identifier only orders items for presentation. This statement concerns ranking results, not the input-index field, which records original provenance.

P3 (fail-closed required data). A missing required metric or non-finite/negative uncertainty is assessed as a failure with infinite violation. It cannot pass by being silently dropped or coerced to zero.

P4 (point-estimate reduction). At $z = 0$, $B_{ij}^{-} = B_{ij}^{+} = \bar{x}_{ij}$, so the physical feasibility indicator and aggregate violation equal the point-estimate rule. Among candidates that are equally feasible under the QUBO layer, the ordering first prefers physically feasible candidates and then lower aggregate violation, consistent with Deb's feasibility-first comparison [9]. This is a reduction of the physical ordering, not a claim that all details of Deb's evolutionary algorithm are reproduced.

The regression tests exercise monotonicity, permutation-invariant tie groups, fail-closed missing/non-finite values, and the $z = 0$ equivalence between point-estimate and CB-PFR modes.

E. Archived Campaigns and Exploratory Limits

The six-layout campaign contains 18 recorded multigroup OpenMC runs: six labels across seed rounds 820000, 830000, and 840000. Each used the documented seven-group C5G7 energy structure, 7,000 total batches, 500 inactive batches, 10,000 particles per batch, and two threads. The records contain unique seeds and statepoint hashes and a common multigroup cross-section (MGXS) hash. These archive checks support traceability, but the model geometry/material mapping and spatial-tally correspondence have not been independently reviewed against the primary specification.

A separate three-seed UMMU repeatability assessment used the same total batch count (7,000) and particles per batch (10,000) under two protocols: 500 inactive batches in one archived run set and 100 inactive batches in a later run set. The 500-inactive simulations completed and their raw HDF5 audits passed, but the initial aggregation job stopped because the source-convergence report for seed 720000 was absent. The archived statepoints were subsequently inspected directly for active-batch entropy and k-generation traces. The later 100-inactive run set produced these reports but logged repeated warnings that fission-source sites fell outside the entropy mesh. The two protocols are therefore reported separately rather than pooled or presented as interchangeable replications.

No design-specific acceptance rule was independently approved before the six-layout campaign. Therefore, $k_{\text{eff}} \in [1.14, 1.21]$, $F_{\max} = 1.35$, $z = 1$, scales $d_k = 0.01$, $d_F = 0.1$, and unit weights are explicitly retrospective illustrative settings. They exercise the software pathway only. A sweep over $F_{\max} \in \{1.30, 1.32, 1.35, 1.80, 1.82, 1.87, 1.90\}$ and $z \in \{0, 1, 2\}$ is sensitivity analysis, not confirmation.

III. Results

A. Symmetry-Class Aggregation

Table I summarizes class-level run results. $s_{k,\mathrm{run}}$ is the sample standard deviation of run-level k_{eff} means within each class; $\bar{s}_{k,\mathrm{within}}$ is the mean reported within-run standard deviation. For the illustrative CB-PFR screen, $s_k^* = \max(s_{k,\mathrm{run}}, \bar{s}_{k,\mathrm{within}})$. The s_F column is the sample standard deviation of run-level F observations pooled within the class. Values are descriptive, not estimates of calibrated confidence coverage.

The mirror-pair intervals include zero and are wide relative to the point differences. No equivalence margin was prospectively specified before these data were examined; therefore, this is a descriptive check, not a formal equivalence test. The theoretical symmetry reduction follows from the declared geometry mapping and symmetry assumptions, while these $n = 3$-per-label intervals do not independently validate the numerical equivalence of the run outcomes.

The measured F means separate the four classes descriptively, while mirror members within S1 and S4 should not be assigned distinct physical ranks from their small sampling differences. Across the 18 layout runs, first-to-last active-quartile mean shifts ranged from -1.81×10^{-3} to $+3.11 \times 10^{-3}$ for source entropy and from -1.07×10^{-3} to $+1.16 \times 10^{-3}$ for k-generation. These summaries are diagnostic only: no convergence tolerance was set prospectively, and small quartile shifts do not certify source convergence.

B. CB-PFR Status and Threshold Sensitivity

For $z = 1$, the illustrative k_{eff} bounds all fall within $[1.14, 1.21]$, so $V_k = 0$ for every class. Consequently, the decision in this case study is determined only by the F constraint; the manuscript does not present k_{eff} and F as two active filters in this particular screen. Table III reports $B_k^{\pm}$, B_F^{+}, metric-specific violations, their total, and the rank/tie group. S1 and S2 are tied at zero violation; neither is preferred without a declared secondary objective. S4 has lower violation than S3 among failing classes. Since $F_{\max}$ was selected retrospectively, these statuses demonstrate software behavior only.

Table IV reports counts as *label strings / symmetry classes*; for example, S1 contributes two label strings but one physical class. The $z = 0$ and $z = 1$ counts coincide across this tested grid; $z = 2$ changes the count only at $F_{\max} = 1.87$. The passing count is therefore dependent on the retrospective cutoff, which is not independently justified as a physical limit.

C. UMMU Repeatability and Protocol Audit

Table V compares two archived three-seed UMMU run sets. Both used 7,000 total batches, 10,000 particles per batch, the same multigroup cross-section hash, unique random seeds, and passing raw HDF5/group-structure audits; the inactive-batch setting differs. The mean shifts from $k_{\mathrm{ref}} = 1.18655$ are expressed in absolute pcm as $(\bar{k} - k_{\mathrm{ref}}) \times 10^5$. The project reference uncertainty is $u_{\mathrm{ref}} = 9.5 \times 10^{-5}$ in absolute k_{eff} units (9.5 pcm); it is not a relative uncertainty or an official benchmark tolerance.

For a descriptive combined-uncertainty score only, we use

$$u_{\mathrm{run}} = \frac{\max(s_{\mathrm{between}}, s_{\mathrm{within}})}{\sqrt{n}}, \qquad z_{\mathrm{comb}} = \frac{\bar{k} - k_{\mathrm{ref}}}{\sqrt{u_{\mathrm{ref}}^2 + u_{\mathrm{run}}^2}},$$

where s_{between} is the sample SD of run means and s_{within} is the pooled within-run reported SD. This conservative operational convention yields $z_{\mathrm{comb}} = -0.434$ for the 500-inactive set and -0.698 for the 100-inactive set. It is a descriptive project calculation, not an acceptance criterion or evidence of benchmark validation.

The two protocol summaries must not be merged: between-run SD is about 1.63 times the pooled within-run SD for Protocol A but only 0.081 times that SD for Protocol B. The latter is a pronounced under-dispersion relative to the run-level uncertainty estimates and requires investigation rather than interpretation as unusually strong repeatability. Active-trace first-to-last-quartile k-generation shifts are $+24.0$ to $+65.1$ pcm for Protocol A and -16.1 to $+23.5$ pcm for Protocol B. The corresponding entropy shifts are -0.00195 to -0.00007 and -1.456 to -0.475, respectively; the 100-inactive logs also recorded repeated source sites outside the entropy mesh. These traces do not certify source convergence. Neither protocol is treated as a validated canonical estimate; model/tally review and convergence investigation remain necessary.

IV. Discussion

The results illustrate three distinct issues for a decision-support workflow. First, the composition-only QUBO enforces the material count but ties all feasible layouts by construction. Second, explicit uncertainty bounds, fail-closed data handling, and stable tie groups make the reasons for a screening decision inspectable; they do not calibrate the bounds or validate the acceptance thresholds. Third, the mirror-pair intervals are too wide to establish numerical equivalence from the available three replicates per label. The F means separate the four class means descriptively, but this small enumerated case does not demonstrate comparative benefit over a baseline or the value of running an optimizer before simulation.

The proxy $F = \max(R_i)/\overline{R}$ is itself a statistic built from noisy tally estimates; taking a maximum can introduce upward selection bias, particularly for shorter runs. The reported s_F values are between-run sample standard deviations of the layout-level proxy, not within-run standard errors of the maximum or calibrated uncertainty bounds. Correlations between regional tallies are not modeled separately here. The paper therefore does not interpret s_F as validated uncertainty coverage.

The paper does not claim that the selected proxy is an engineering acceptance criterion. A stronger confirmatory study would require physical limits justified independently of this campaign, a prospectively specified analysis, independent model/tally review, and an adequate comparison against simple methods. A larger candidate space and a meaningful physics-sensitive pre-physics objective would be

TABLE I
SIX-LAYOUT MULTIGROUP OPENMC CAMPAIGN AGGREGATED BY DIAGONAL-REFLECTION CLASS.

Class	Layout labels	n	Mean k_{eff}	$s_{k,\text{run}}$	$\bar{s}_{k,\text{within}}$	Mean F	s_F
S1	MMUU / MUMU	6	1.143686	7.10×10^{-5}	9.54×10^{-5}	1.295867	0.000960
S2	MUUM	3	1.156263	8.59×10^{-5}	9.53×10^{-5}	1.316149	0.001297
S3	UMMU	3	1.186499	1.40×10^{-4}	9.70×10^{-5}	1.867100	0.001541
S4	UMUM / UUMM	6	1.204665	7.94×10^{-5}	9.71×10^{-5}	1.816406	0.000673

TABLE II
EXPLORATORY DIFFERENCES BETWEEN LAYOUT LABELS RELATED BY THE DECLARED DIAGONAL REFLECTION. DIFFERENCES ARE FIRST LABEL MINUS SECOND LABEL; INTERVALS ARE TWO-SIDED 95% WELCH INTERVALS FROM THREE RUNS PER LABEL.

Pair	Metric	Mean difference	95% interval lower	95% interval upper
MMUU − MUMU	k_{eff} (pcm)	−2.81	−22.88	17.26
MMUU − MUMU	F	−0.000995	−0.003152	0.001162
UMUM − UUMM	k_{eff} (pcm)	4.67	−16.52	25.86
UMUM − UUMM	F	0.000628	−0.001469	0.002726

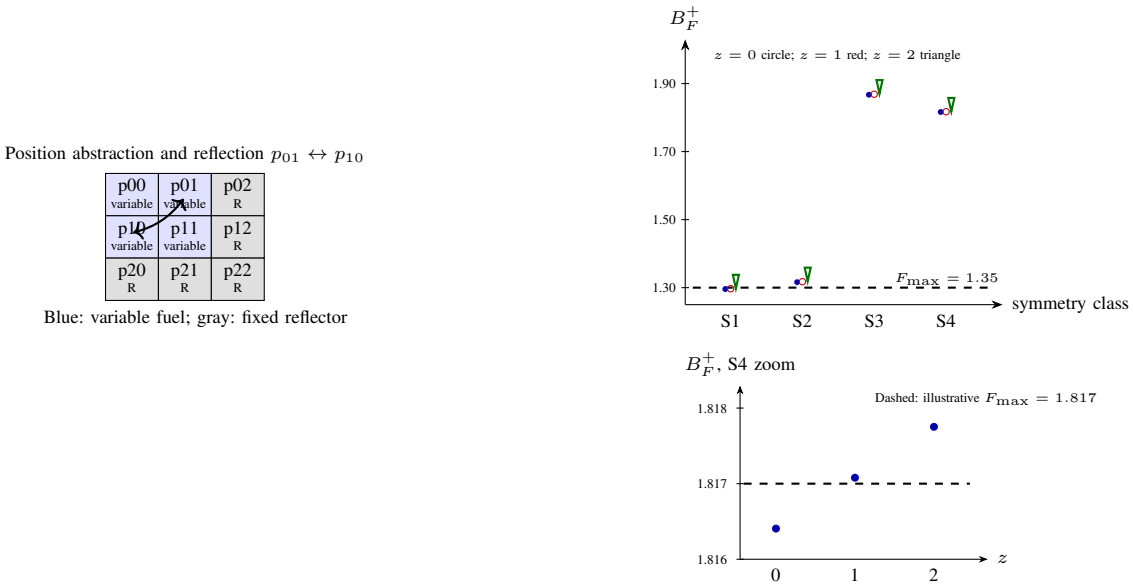


Fig. 1. (Left) Four variable fuel positions and five fixed reflector positions; diagonal reflection exchanges p_{01} and p_{10}. (Right, upper) Class-level operational upper bounds B_F^+ for $z = 0, 1, 2$ with the study's retrospective $F_{\max} = 1.35$. (Right, lower) A deliberately zoomed illustration of the S4 bound crossing a separate illustrative $F_{\max} = 1.817$ as z increases. The zoom threshold is shown only to explain the uncertainty-bound mechanism; it is not an acceptance limit used in the results.

required to evaluate claims about optimizer utility. Constraint-handling methods and simulation ranking-and-selection have established literature [9], [14]. For uncertainty terminology, the paper distinguishes within-run Monte Carlo standard uncertainties from observed between-run dispersion, consistent with measurement-uncertainty guidance [15]. Other C5G7 deterministic transport studies provide methodological context but are not treated as direct performance baselines for CB-PFR [16]. The contribution is the explicit software workflow and its evidence-bounded audit, not priority for feasibility-first ranking.

V. CONCLUSION

This paper reports an exploratory, evidence-audited CB-PFR workflow on six enumerated C5G7-derived fuel layouts. The composition QUBO assigns zero energy to each feasible layout; diagonal-reflection symmetry reduces six labels to four classes. Under a retrospective illustrative screen, two classes pass and two fail, while the sensitivity table makes threshold dependence explicit. The separate UMMU run sets give absolute reference differences of -5.68 pcm and -7.64 pcm under 500- and 100-inactive-batch protocols, respectively. Their repeatability diagnostics differ materially, and neither establishes benchmark acceptance. The evidence supports workflow traceability and transparent handling of ties, not physical optimality, calibrated confidence coverage, full validation, or method superiority.

TABLE III
ILLUSTRATIVE CLASS-LEVEL CB-PFR SCREEN: $k \in [1.14, 1.21]$, $F_{\max} = 1.35$, $z = 1$; RETROSPECTIVE LIMITS ONLY.

Class	B_k^-	B_k^+	B_F^+	V_k	V_F	V_{total}	Status	Rank / tie-group
S1	1.143591	1.143781	1.296827	0.0000	0.0000	0.0000	Pass	Tie 1
S2	1.156168	1.156358	1.317446	0.0000	0.0000	0.0000	Pass	Tie 1
S3	1.186359	1.186639	1.868641	0.0000	5.1864	5.1864	Fail	Rank 3
S4	1.204568	1.204762	1.817079	0.0000	4.6708	4.6708	Fail	Rank 2

TABLE IV
RETROSPECTIVE SENSITIVITY SWEEP: PASSING LABEL STRINGS / SYMMETRY CLASSES.

$F_{\max}$	$z = 0$	$z = 1$	$z = 2$
1.30	2 / 1	2 / 1	2 / 1
1.32	3 / 2	3 / 2	3 / 2
1.35	3 / 2	3 / 2	3 / 2
1.80	3 / 2	3 / 2	3 / 2
1.82	5 / 3	5 / 3	5 / 3
1.87	6 / 4	6 / 4	5 / 3
1.90	6 / 4	6 / 4	6 / 4

SOFTWARE AND DATA AVAILABILITY

The accompanying anonymous-review supplement contains the 18-run six-layout table, six UMMU run-level records with statepoint hashes, a two-protocol summary and source-trace diagnostics, and the scripts needed to reproduce the reported aggregation rules. The project identity and permanent public release are withheld from this double-blind manuscript; no DOI or public release is claimed in this review version.

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TABLE V
SEPARATE UMMU RUN PROTOCOLS AND REPEATABILITY SUMMARIES.

Protocol	Inactive batches	Mean k_{eff}	Δk (pcm)	Between-run SD (pcm)	Pooled within-run SD (pcm)
A	500	1.18649321	-5.68	15.57	9.58
B	100	1.18647358	-7.64	0.77	9.45